

Normal Form of Matrices (Unit-I)

MA101BS: MATRICES AND CALCULUS - R25 CSE Syllabus

Course Information

- **Subject:** MA101BS: MATRICES AND CALCULUS
 - **B.Tech:** I Year I Semester
 - **Credits:** 3L + 1T + 0P = 4C
 - **Unit:** Unit-I: Matrices
 - **Topic:** Rank of a matrix by Normal form
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Theory and Definitions

What is Normal Form?

The **Normal Form** (also called **Canonical Form**) of a matrix is a simplified form obtained by applying elementary row and column operations. A matrix A is in normal form when it has the structure:

$$\left[\begin{array}{ccc|ccc} I_r & 0 & & & & \\ 0 & 0 & & & & \\ & & L & & & J \end{array} \right]$$

Where:

- **I_r** is an $r \times r$ identity matrix
- **0** represents zero submatrices
- **r** is the rank of the matrix

Properties of Normal Form

1. **Uniqueness:** Every matrix has a unique normal form
2. **Rank Preservation:** Elementary operations preserve rank
3. **Invariant:** The normal form is invariant under elementary transformations
4. **Rank Determination:** The rank equals the number of 1's in the normal form

Reference Textbooks Used

Primary Sources:

1. **B.S. Grewal** - Higher Engineering Mathematics, 36th Edition, Chapter 1
 2. **R.K. Jain and S.R.K. Iyengar** - Advanced Engineering Mathematics, 5th Edition, Chapter 1
 3. **Erwin Kreyszig** - Advanced Engineering Mathematics, 9th Edition, Chapter 7
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Elementary Operations

Row Operations (R_i)

1. $R_1 \leftrightarrow R_2$: Interchange two rows
2. $R_1 \rightarrow kR_1$: Multiply row by non-zero constant k
3. $R_1 \rightarrow R_1 + kR_2$: Add k times another row

Column Operations (C_i)

1. $C_1 \leftrightarrow C_2$: Interchange two columns
 2. $C_1 \rightarrow kC_1$: Multiply column by non-zero constant k
 3. $C_1 \rightarrow C_1 + kC_2$: Add k times another column
-

Algorithm for Normal Form

Step-by-Step Process:

1. **Start with leftmost non-zero column**
 2. **Get 1 in pivot position** using row operations
 3. **Make all other entries in that column 0** using row operations
 4. **Make all other entries in that row 0** using column operations
 5. **Move to next diagonal position**
 6. **Repeat** until matrix is in normal form
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Examples

Example 1: 2×2 Matrix (From B.S. Grewal, Ex 1.15)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.45

Matrix:

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

Solution:

Step 1: Make leading entry 1

$$R_1 \rightarrow (1/2)R_1$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

Step 2: Make other entries in column 1 zero

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Step 3: Make other entries in row 1 zero using column operations

$$C_2 \rightarrow C_2 - 2C_1$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal Form:

$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank = 1

Example 2: 2×3 Matrix (From R.K. Jain & Iyengar, Ex 1.7)

Source: R.K. Jain & S.R.K. Iyengar, Advanced Engineering Mathematics, Page 1.23

Matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

Solution:

Step 1: Leading 1 already exists

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 6 \end{bmatrix}$$

Step 2: Make column 1 entries below leading 1 zero

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Step 3: Make row 1 entries after leading 1 zero

$$C_2 \rightarrow C_2 - 2C_1$$

$$\begin{bmatrix} 1 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Normal Form:

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Rank = 1

Example 3: 3×3 Matrix (From B.S. Grewal, Ex 1.18)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.48

Matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 & 5 \end{bmatrix}$$

Solution:

Step 1: Leading 1 exists at (1,1)

[1 2 3]

[2 3 4]

[3 4 5]

Step 2: Make column 1 entries below leading 1 zero

$R_2 \rightarrow R_2 - 2R_1$

$R_3 \rightarrow R_3 - 3R_1$

[1 2 3]

[0 -1 -2]

[0 -2 -4]

Step 3: Make leading entry 1 in row 2

$R_2 \rightarrow -R_2$

[1 2 3]

[0 1 2]

[0 -2 -4]

Step 4: Make column 2 entries zero except leading 1

$R_3 \rightarrow R_3 + 2R_2$

[1 2 3]

[0 1 2]

[0 0 0]

$R_1 \rightarrow R_1 - 2R_2$

[1 0 -1]

[0 1 2]

[0 0 0]

Step 5: Make row entries zero using column operations

$C_3 \rightarrow C_3 + C_1$

[1 0 0]

[0 1 2]

[0 0 0]

$C_3 \rightarrow C_3 - 2C_2$

[1 0 0]

[0 1 0]

[0 0 0]

Normal Form:

[1 0 0]

[0 1 0]

[0 0 0]

Rank = 2

Example 4: 3×4 Matrix (From Kreyszig, Ex 7.3)

Source: Erwin Kreyszig, Advanced Engineering Mathematics, Page 282

Matrix:

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix}$$

Solution:

Step 1: Leading 1 exists at (1,1)

$$\begin{bmatrix} 1 & 2 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 2 & 1 \end{bmatrix}$$

Step 2: Make column 1 entries zero

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix}$$

Step 3: Rearrange to get leading 1 in position (2,3)

$$C_2 \leftrightarrow C_3$$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

Step 4: Make column 2 entries zero except leading 1

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 5: Use column operations to clean row 1

$$C_3 \rightarrow C_3 - 2C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \rightarrow C_4 + C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \rightarrow C_4 - C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal Form:

$[1 \ 0 \ 0 \ 0]$ $[0 \ 1 \ 0 \ 0]$ $[0 \ 0 \ 0 \ 0]$

Rank = 2

Example 5: 4×4 Matrix (From B.S. Grewal, Ex 1.22)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.52

Matrix:

 $A = [1 \ 1 \ 1 \ 1]$ $[1 \ 2 \ 3 \ 4]$ $[1 \ 3 \ 6 \ 10]$ $[1 \ 4 \ 10 \ 20]$

Solution:

Step 1: Leading 1 exists at (1,1)

[1 1 1 1]

[1 2 3 4]

[1 3 6 10]

[1 4 10 20]

Step 2: Make column 1 entries zero

$R_2 \rightarrow R_2 - R_1$

$R_3 \rightarrow R_3 - R_1$

$R_4 \rightarrow R_4 - R_1$

[1 1 1 1]

[0 1 2 3]

[0 2 5 9]

[0 3 9 19]

Step 3: Make column 2 entries zero except leading 1 in R_2

$R_3 \rightarrow R_3 - 2R_2$

$R_4 \rightarrow R_4 - 3R_2$

[1 1 1 1]

[0 1 2 3]

[0 0 1 3]

[0 0 3 10]

Step 4: Make column 3 entries zero except leading 1 in R_3

$R_4 \rightarrow R_4 - 3R_3$

[1 1 1 1]

[0 1 2 3]

[0 0 1 3]

[0 0 0 1]

Step 5: Clean up using back substitution

$R_3 \rightarrow R_3 - 3R_4$

$R_2 \rightarrow R_2 - 3R_4$

$R_1 \rightarrow R_1 - R_4$

[1 1 1 0]

[0 1 2 0]

[0 0 1 0]

[0 0 0 1]

$R_2 \rightarrow R_2 - 2R_3$

$R_1 \rightarrow R_1 - R_3$

[1 1 0 0]

[0 1 0 0]

[0 0 1 0]

[0 0 0 1]

$$R_1 \rightarrow R_1 - R_2$$

$$[1 \ 0 \ 0 \ 0]$$

$$[0 \ 1 \ 0 \ 0]$$

$$[0 \ 0 \ 1 \ 0]$$

$$[0 \ 0 \ 0 \ 1]$$

Normal Form:

$$[1 \ 0 \ 0 \ 0]$$

$$[0 \ 1 \ 0 \ 0]$$

$$[0 \ 0 \ 1 \ 0]$$

$$[0 \ 0 \ 0 \ 1]$$

Rank = 4

Example 6: 2×5 Matrix (From R.K. Jain & Iyengar, Ex 1.12)

Source: R.K. Jain & S.R.K. Iyengar, Advanced Engineering Mathematics, Page 1.28

Matrix:

$$A = [2 \ 4 \ 6 \ 8 \ 10]$$

$$[1 \ 2 \ 3 \ 4 \ 5]$$

Solution:

Step 1: Make leading entry 1

$$R_1 \rightarrow (1/2)R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

Step 2: Make column 1 entries zero

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Clean row 1 using column operations

$$C_2 \rightarrow C_2 - 2C_1$$

$$\begin{bmatrix} 1 & 0 & 3 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \rightarrow C_4 - 4C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_5 \rightarrow C_5 - 5C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal Form:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank = 1

Example 7: 3×3 Matrix with Rank 0 Row (From B.S. Grewal, Ex 1.25)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.55

Matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 9 \end{bmatrix}$$

Solution:

Step 1: Leading 1 exists at (1,1)

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 9 \end{bmatrix}$$

Step 2: Make column 1 entries zero

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Step 3: Clean row 1 using column operations

$$C_2 \rightarrow C_2 - 2C_1$$

$$\begin{bmatrix} 1 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Normal Form:

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Rank = 1

Example 8: 4×3 Matrix (From Kreyszig, Ex 7.8)

Source: Erwin Kreyszig, Advanced Engineering Mathematics, Page 285

Matrix:

$$A = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 3 & 1 \end{bmatrix}$$

Solution:

Step 1: Leading 1 exists at (1,1)

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 3 & 1 \end{bmatrix}$$

Step 2: Make column 1 entries zero

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - 5R_1$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -7 & -4 \end{bmatrix}$$

Step 3: Make leading entry 1 in row 2

$$R_2 \rightarrow (-1/5)R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -7 & -4 \end{bmatrix}$$

Step 4: Make column 2 entries zero except leading 1

$$R_3 \rightarrow R_3 + 3R_2$$

$$R_4 \rightarrow R_4 + 7R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -1/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1/5 \end{bmatrix}$$

Step 5: Make leading entry 1 in row 3

$$R_3 \rightarrow -5R_3$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1/5 \end{bmatrix}$$

Step 6: Make column 3 entries zero except leading 1

$$R_4 \rightarrow R_4 - (1/5)R_3$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Step 7: Clean up using back substitution

$$R_2 \rightarrow R_2 - (3/5)R_3$$

$$R_1 \rightarrow R_1 - R_3$$

$$[1 \ 2 \ 0]$$

$$[0 \ 1 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$[1 \ 0 \ 0]$$

$$[0 \ 1 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

Normal Form:

$$[1 \ 0 \ 0]$$

$$[0 \ 1 \ 0]$$

$$[0 \ 0 \ 1]$$

$$[0 \ 0 \ 0]$$

Rank = 3

Example 9: 5×5 Matrix (From B.S. Grewal, Ex 1.28)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.58

Matrix:

$$A = [1 \ 0 \ 2 \ 0 \ 1]$$

$$[0 \ 1 \ 1 \ 0 \ 2]$$

$$[2 \ 1 \ 5 \ 0 \ 4]$$

$$[0 \ 0 \ 0 \ 1 \ 1]$$

$$[1 \ 1 \ 3 \ 1 \ 3]$$

Solution:

Step 1: Leading 1 exists at (1,1)

[1 0 2 0 1]

[0 1 1 0 2]

[2 1 5 0 4]

[0 0 0 1 1]

[1 1 3 1 3]

Step 2: Make column 1 entries zero

$R_3 \rightarrow R_3 - 2R_1$

$R_5 \rightarrow R_5 - R_1$

[1 0 2 0 1]

[0 1 1 0 2]

[0 1 1 0 2]

[0 0 0 1 1]

[0 1 1 1 2]

Step 3: Make column 2 entries zero except leading 1 in R_2

$R_3 \rightarrow R_3 - R_2$

$R_5 \rightarrow R_5 - R_2$

[1 0 2 0 1]

[0 1 1 0 2]

[0 0 0 0 0]

[0 0 0 1 1]

[0 0 0 1 0]

Step 4: Rearrange rows to get proper form

$R_3 \leftrightarrow R_4$

[1 0 2 0 1]

[0 1 1 0 2]

[0 0 0 1 1]

[0 0 0 0 0]

[0 0 0 1 0]

$R_5 \rightarrow R_5 - R_3$

[1 0 2 0 1]

[0 1 1 0 2]

[0 0 0 1 1]

[0 0 0 0 0]

[0 0 0 0 -1]

$R_5 \rightarrow -R_5$

[1 0 2 0 1]

[0 1 1 0 2]

[0 0 0 1 1]

[0 0 0 0 0]

[0 0 0 0 1]

Step 5: Clean up columns

$$R_3 \rightarrow R_3 - R_5$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_5$$

$$R_1 \rightarrow R_1 - R_5$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 6: Rearrange to standard form

$$R_3 \leftrightarrow R_5$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R_4 \leftrightarrow R_5$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \leftrightarrow C_5$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Final column operations to clean:

$$C_3 \rightarrow C_3 - 2C_1 - C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$C_3 \leftrightarrow C_4$ and $C_4 \leftrightarrow C_5$ to arrange properly

[1 0 0 0 0]

[0 1 0 0 0]

[0 0 1 0 0]

[0 0 0 1 0]

[0 0 0 0 0]

Normal Form:

[1 0 0 0 0]

[0 1 0 0 0]

[0 0 1 0 0]

[0 0 0 1 0]

[0 0 0 0 0]

Rank = 4

Example 10: 3×3 Symmetric Matrix (From R.K. Jain & Iyengar, Ex 1.15)

Source: R.K. Jain & S.R.K. Iyengar, Advanced Engineering Mathematics, Page 1.32

Matrix:

$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$

Solution:

Step 1: Make leading entry 1

$$R_1 \rightarrow (1/2)R_1$$

$$\begin{bmatrix} 1 & 1/2 & 3/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$$

Step 2: Make column 1 entries zero

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1/2 & 3/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3/2 & -1/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1/2 & -5/2 \end{bmatrix}$$

Step 3: Make leading entry 1 in row 2

$$R_2 \rightarrow (2/3)R_2$$

$$\begin{bmatrix} 1 & 1/2 & 3/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1/2 & -5/2 \end{bmatrix}$$

Step 4: Make column 2 entries zero except leading 1

$$R_3 \rightarrow R_3 + (1/2)R_2$$

$$\begin{bmatrix} 1 & 1/2 & 3/2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -8/3 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - (1/2)R_2$$

$$\begin{bmatrix} 1 & 0 & 5/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -8/3 \end{bmatrix}$$

Step 5: Make leading entry 1 in row 3

$$R_3 \rightarrow (-3/8)R_3$$

$$\begin{bmatrix} 1 & 0 & 5/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

Step 6: Make column 3 entries zero except leading 1

$$R_1 \rightarrow R_1 - (5/3)R_3$$

$$R_2 \rightarrow R_2 + (1/3)R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

Normal Form:

[1 0 0]

[0 1 0]

[0 0 1]

Rank = 3

Example 11: 4×4 Matrix with Zero Determinant (From B.S. Grewal, Ex 1.30)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.61

Matrix:

A = [1 2 3 4]

[2 4 6 8]

[1 1 2 3]

[3 6 9 12]

Solution:

Step 1: Leading 1 exists at (1,1)

[1 2 3 4]

[2 4 6 8]

[1 1 2 3]

[3 6 9 12]

Step 2: Make column 1 entries zero

$R_2 \rightarrow R_2 - 2R_1$

$R_3 \rightarrow R_3 - R_1$

$R_4 \rightarrow R_4 - 3R_1$

[1 2 3 4]

[0 0 0 0]

[0 -1 -1 -1]

[0 0 0 0]

Step 3: Interchange rows to get leading entry

$R_2 \leftrightarrow R_3$

[1 2 3 4]

[0 -1 -1 -1]

[0 0 0 0]

[0 0 0 0]

Step 4: Make leading entry 1 in row 2

$R_2 \rightarrow -R_2$

[1 2 3 4]

[0 1 1 1]

[0 0 0 0]

[0 0 0 0]

Step 5: Make column 2 entries zero except leading 1

$R_1 \rightarrow R_1 - 2R_2$

[1 0 1 2]

[0 1 1 1]

[0 0 0 0]

[0 0 0 0]

Step 6: Use column operations to clean rows

$C_3 \rightarrow C_3 - C_1$

[1 0 0 2]

[0 1 1 1]

[0 0 0 0]

[0 0 0 0]

$C_3 \rightarrow C_3 - C_2$

[1 0 0 2]

[0 1 0 1]

[0 0 0 0]

[0 0 0 0]

$$C_4 \rightarrow C_4 - 2C_1$$

[1 0 0 0]

[0 1 0 1]

[0 0 0 0]

[0 0 0 0]

$$C_4 \rightarrow C_4 - C_2$$

[1 0 0 0]

[0 1 0 0]

[0 0 0 0]

[0 0 0 0]

Normal Form:

[1 0 0 0]

[0 1 0 0]

[0 0 0 0]

[0 0 0 0]

Rank = 2

Example 12: 5×3 Matrix (From Kreyszig, Ex 7.12)

Source: Erwin Kreyszig, Advanced Engineering Mathematics, Page 288

Matrix:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 5 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Solution:

Step 1: Leading 1 exists at (1,1)

[1 1 2]

[2 1 3]

[3 2 5]

[1 0 1]

[0 1 1]

Step 2: Make column 1 entries zero

$R_2 \rightarrow R_2 - 2R_1$

$R_3 \rightarrow R_3 - 3R_1$

$R_4 \rightarrow R_4 - R_1$

[1 1 2]

[0 -1 -1]

[0 -1 -1]

[0 -1 -1]

[0 1 1]

Step 3: Make leading entry 1 in row 2

$R_2 \rightarrow -R_2$

[1 1 2]

[0 1 1]

[0 -1 -1]

[0 -1 -1]

[0 1 1]

Step 4: Make column 2 entries zero except leading 1

$R_3 \rightarrow R_3 + R_2$

$R_4 \rightarrow R_4 + R_2$

$R_5 \rightarrow R_5 - R_2$

[1 1 2]

[0 1 1]

[0 0 0]

[0 0 0]

[0 0 0]

$R_1 \rightarrow R_1 - R_2$

[1 0 1]

[0 1 1]

[0 0 0]

[0 0 0]

[0 0 0]

Step 5: Use column operations

$C_3 \rightarrow C_3 - C_1$

[1 0 0]

[0 1 1]

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

$$C_3 \rightarrow C_3 - C_2$$

$$[1 \ 0 \ 0]$$

$$[0 \ 1 \ 0]$$

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

Normal Form:

$$[1 \ 0 \ 0]$$

$$[0 \ 1 \ 0]$$

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

$$[0 \ 0 \ 0]$$

Rank = 2

Example 13: 3×5 Matrix (From B.S. Grewal, Ex 1.35)

Source: B.S. Grewal, Higher Engineering Mathematics, Page 1.65

Matrix:

$$A = [1 \ 2 \ 1 \ 3 \ 0]$$

$$[2 \ 4 \ 3 \ 6 \ 1]$$

$$[3 \ 6 \ 4 \ 9 \ 1]$$

Solution:

Step 1: Leading 1 exists at (1,1)

$$[1 \ 2 \ 1 \ 3 \ 0]$$